

National Coverage Determination (NCD)

External Counterpulsation (ECP) for Severe Angina

20.20

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Tracking Information

Publication Number

100-3

Manual Section Number

20.20

Manual Section Title

External Counterpulsation (ECP) for Severe Angina

Version Number**Effective Date of this Version**

11/15/2001

Ending Effective Date of this Version

03/20/2006

Implementation Date

11/15/2001

Description Information

Benefit Category

Physicians' Services

Please Note: This may not be an exhaustive list of all applicable Medicare benefit categories for this item or service.

Item/Service Description

External counterpulsation (ECP), commonly referred to as enhanced external counterpulsation, is a non-invasive outpatient treatment for coronary artery disease refractory to medical and/or surgical therapy.

Indications and Limitations of Coverage

Although ECP devices are cleared by the Food and Drug Administration (FDA) for use in treating a variety of cardiac conditions, including stable or unstable angina pectoris, acute myocardial infarction and cardiogenic shock, the use of this device to treat cardiac conditions other than stable angina pectoris is not covered, since only that use has developed sufficient evidence to demonstrate its medical effectiveness. Non-coverage of hydraulic versions of these types of devices remains in force.

Coverage is provided for the use of ECP for patients who have been diagnosed with disabling angina (Class III or Class IV, Canadian Cardiovascular Society Classification or equivalent classification) who, in the opinion of a cardiologist or cardiothoracic surgeon, are not readily amenable to surgical intervention, such as PTCA or cardiac bypass because:

1. Their condition is inoperable, or at high risk of operative complications or post-operative failure;
2. Their coronary anatomy is not readily amenable to such procedures; or
3. They have co-morbid states which create excessive risk.

A full course of therapy usually consists of 35 one-hour treatments, which may be offered once or twice daily, usually 5 days per week. The patient is placed on a treatment table where their lower trunk and lower extremities are wrapped in a series of three compressive air cuffs which inflate and deflate in synchronization with the patient's cardiac cycle.

During diastole the three sets of air cuffs are inflated sequentially (distal to proximal) compressing the vascular beds within the muscles of the calves, lower thighs and upper thighs. This action results in an increase in diastolic pressure, generation of retrograde arterial blood flow and an increase in venous return. The cuffs are deflated simultaneously just prior to systole, which produces a rapid drop in vascular impedance, a decrease in ventricular workload and an increase in cardiac output.

The augmented diastolic pressure and retrograde aortic flow appear to improve myocardial perfusion, while systolic unloading appears to reduce cardiac workload and oxygen requirements. The increased venous return coupled with enhanced systolic flow appears to increase cardiac output. As a result of this treatment, most patients experience increased time until onset of ischemia, increased exercise tolerance, and a reduction in the number and severity of anginal episodes. Evidence was presented that this effect lasted well beyond the

immediate post-treatment phase, with patients symptom-free for several months to two years.

This procedure must be done under direct supervision of a physician.

Claims Processing Instructions

[TN 779 \(Medicare Hospital Manual\)](#)

[TN 1845 \(Medicare Intermediary Manual\)](#)

[TN 748 \(Medicare Hospital Manual\)](#)

[TN 1789 \(Medicare Intermediary Manual\)](#)

[TN 1659 \(Medicare Carriers Manual\)](#)

Transmittal Information

Transmittal Number

146

Coverage Transmittal Link

<https://www.cms.gov/Regulations-and-Guidance/Guidance/Transmittals/Downloads/R146CIM.pdf>

Revision History

10/2001 - Amended to indicate that policy only pertains to ECP devices intended for treatment of cardiac conditions. Effective and implementation dates 11/15/2001. ([TN 146](#)) (CR 1884)

02/2000 - Changed acronym from EECP to ECP, removed requirement limiting coverage to specific ECP systems, and changed CPT code. Effective and implementation dates 04/01/2000. ([TN 122](#)) (CR 1087)

07/1999 - Changed CPT code. Effective date 07/01/1999. (TN 118)

04/1999 - Revised existing noncoverage policy to limited coverage for use in patients with stable angina pectoris and designated CPT code for billing. Effective date 07/01/1999. (TN 111)

National Coverage Analyses (NCAs)

This NCD has been or is currently being reviewed under the National Coverage Determination process. The following are existing associations with NCAs, from the National Coverage Analyses database.

- [Original Consideration for External Counterpulsation \(ECP\) Therapy \(CAG-00002N\)](#) 
- [First reconsideration for External Counterpulsation \(ECP\) Therapy \(CAG-00002R\)](#) 
- [Second reconsideration for External Counterpulsation \(ECP\) Therapy \(CAG-00002R2\)](#) 

Additional Information

Other Versions

Title	Version	Effective Between	
External Counterpulsation (ECP) Therapy for Severe Angina	2	03/20/2006 - N/A	View
External Counterpulsation (ECP) for Severe Angina	1	11/15/2001 - 03/20/2006	You are here